Lab 2 — Agile Backlog Creation & Sprint Simulation in Jira

Course: Software Engineering, PES University (Dept. of CSE)

Problem Statement #46 — Developer Tools & IT Operations: **Remote Team Time-Tracking & Project Approver**

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Objective. Turn the Lab 1 functional requirements into an Agile backlog (Epics → User Stories), prioritise it, estimate it with story points on the Fibonacci scale using Planning Poker, run one-week sprints on a Scrum board (To Do → In Progress → Done) and analyse progress with burndown charts. All work below was done in the live Jira workspace; screenshots of that workspace are embedded in each section.

1. Source requirements (Lab 1)

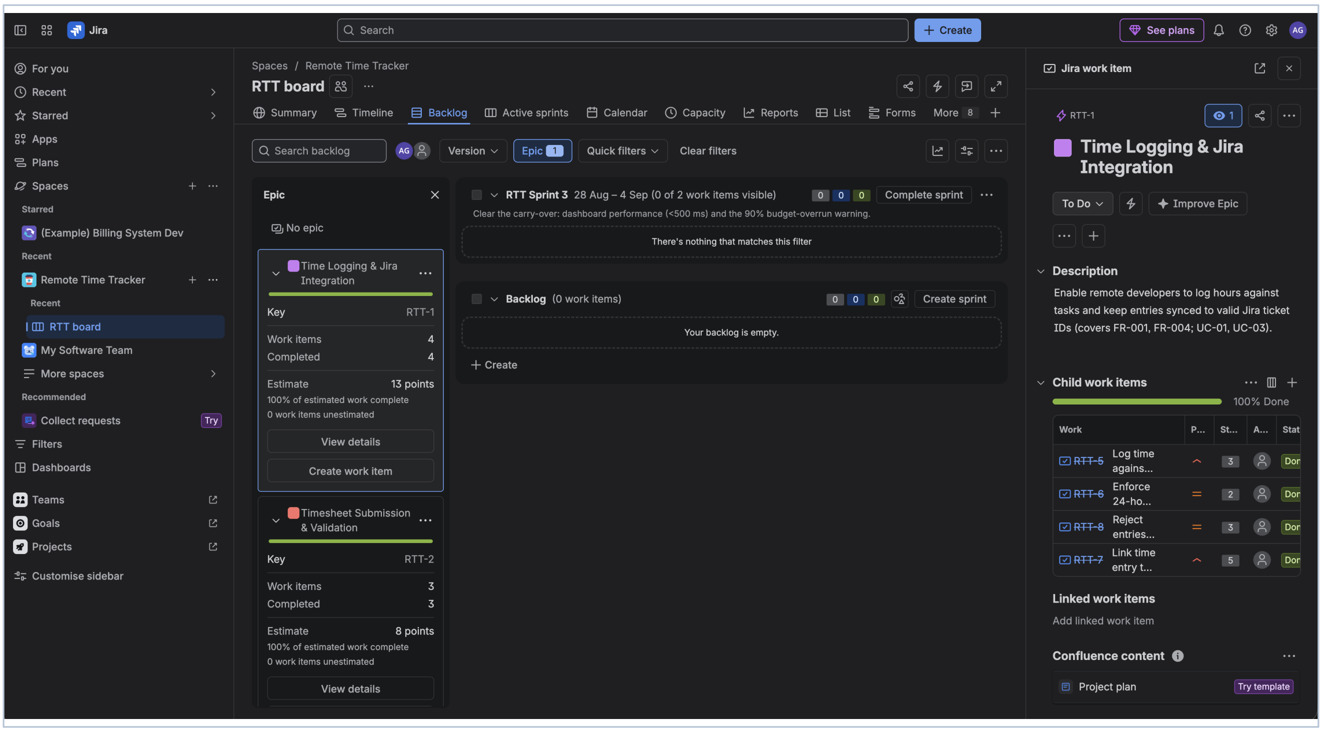
The backlog is derived directly from the Lab 1 requirements table and use-case model for the engineering-productivity platform (developers log hours against Jira tickets, managers approve weekly timesheets, project budget burn-rate is monitored).

Req	Type	Pr.	Requirement (summary)	Realised by
FR-001	F	High	Log time against tasks; block any day's total > 24 h	RTT-5, RTT-6
FR-002	F	High	Developer submits a weekly timesheet for approval	RTT-9, RTT-10, RTT-11
FR-003	F	High	Manager approves/rejects; rejection needs a comment	RTT-12, RTT-13, RTT-14
FR-004	F	Med	Entries synced to Jira ticket IDs; bad tickets rejected	RTT-7, RTT-8
FR-005	F	High	Compute & display real-time project budget burn rate	RTT-16
NFR-001	NF	High	Dashboards update within 500 ms at peak load	RTT-17
NFR-002	NF	High	RBAC — developer cannot approve own timesheet	RTT-15

2. Epics

Four Epics group the requirements into major functional themes. Each Epic is a body of work broken down into User Stories.

Epic	Name	Description	Requirements	Size
RTT-1	**Time Logging & Jira Integration**	Enable remote developers to log hours against tasks and keep entries synced to valid Jira ticket IDs.	FR-001, FR-004 (UC-01, UC-03)	4 / 13 SP
RTT-2	**Timesheet Submission & Validation**	Let developers assemble, validate and submit weekly timesheets, and revise rejected ones.	FR-002 (UC-02, UC-04)	3 / 8 SP
RTT-3	**Managerial Timesheet Approval**	Let managers review, approve or reject timesheets with mandatory rejection comments, enforced by RBAC.	FR-003, NFR-002 (UC-05, UC-06)	4 / 13 SP
RTT-4	**Project Budget Monitoring**	Give managers a real-time budget burn-rate dashboard (<500 ms at peak) that flags >90% projected burn.	FR-005, NFR-001 (UC-07, UC-08)	3 / 18 SP



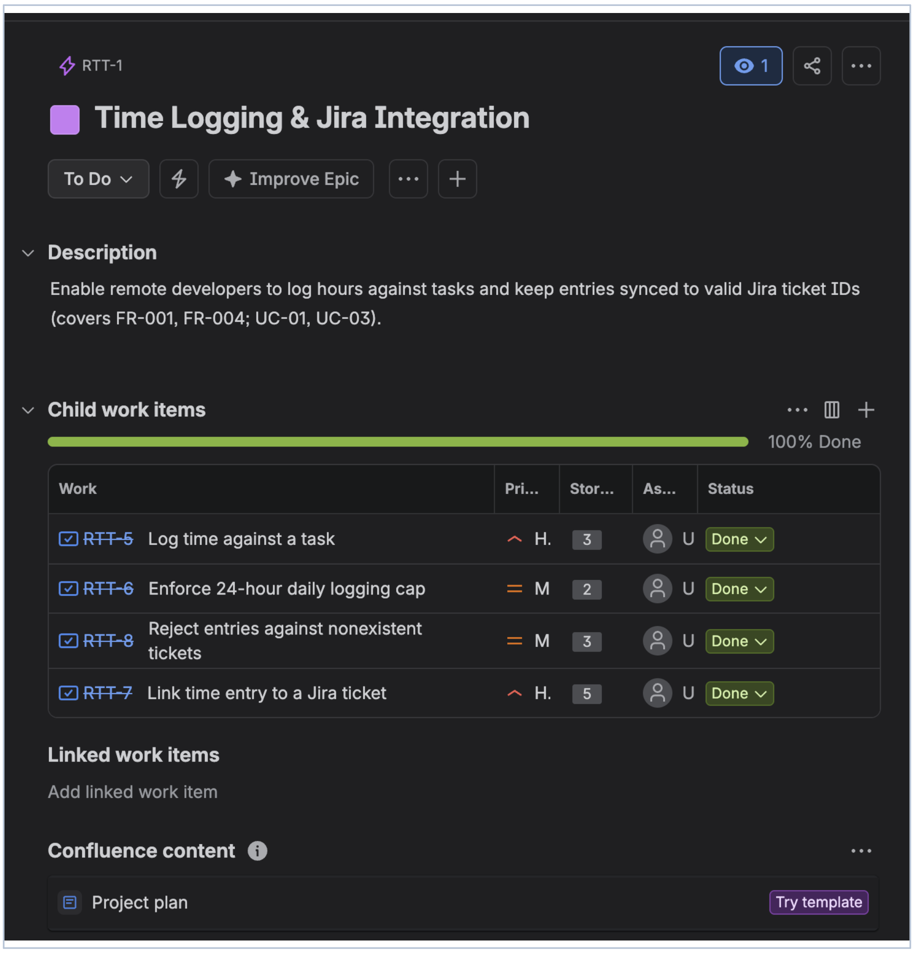
Jira Backlog with the Epic panel open (the 4 Epics RTT-1..RTT-4 with keys, estimates and completion), and Epic RTT-1 opened showing its description and child User Stories.

3. Product backlog — User Stories

Every story uses the template *“As a [role], I want [goal], so that [benefit]”*. Priority (High / Medium / Low) is set by user impact; story points use the Fibonacci scale. “Sprint” and “Final status” record the outcome of the sprint simulation.

ID	Epic	User story	Pr.	SP	Sprint	Status
RTT-5	RTT-1	As a remote developer, I want to log hours worked against a specific task, so that my effort is recorded accurately.	High	3	1	Done
RTT-6	RTT-1	As a remote developer, I want the system to block any single day's logged hours from exceeding 24, so that impossible entries are prevented.	Medium	2	1	Done
RTT-7	RTT-1	As a remote developer, I want each time entry linked to a Jira ticket ID and synced to that ticket, so that my logged work maps to real project tickets.	High	5	1	Done
RTT-8	RTT-1	As a remote developer, I want time entries against nonexistent Jira tickets rejected when I save them, so that the time data stays clean and reliable.	Medium	3	1	Done
RTT-9	RTT-2	As a remote developer, I want to submit my completed weekly timesheet to my engineering manager, so that it can be reviewed and approved.	High	3	1	Done
RTT-10	RTT-2	As a remote developer, I want my timesheet entries validated when I submit, so that errors are caught before the manager review.	Medium	3	1→2	Done (S2)
RTT-11	RTT-2	As a remote developer, I want to edit and resubmit a rejected timesheet, so that I can correct the flagged issues without starting over.	Medium	2	1→2	Done (S2)
RTT-12	RTT-3	As an engineering manager, I want to see all timesheets awaiting my approval in one place, so that I can review them efficiently.	High	2	2	Done
RTT-13	RTT-3	As an engineering manager, I want to approve a reviewed timesheet, so that the developer's hours are confirmed and locked for billing.	High	3	2	Done
RTT-14	RTT-3	As an engineering manager, I want rejection to require a comment, so that the developer understands why it was rejected and what to fix.	High	3	2	Done
RTT-15	RTT-3	As an engineering manager, I want the system to stop any developer from approving or rejecting their own timesheet, so that separation-of-duties controls are enforced.	High	5	2	Done
RTT-16	RTT-4	As an engineering manager, I want a dashboard showing the project's real-time budget burn rate, so that I can monitor spend against the approved budget.	High	5	2	Done
RTT-17	RTT-4	As an engineering manager, I want approval-status and budget dashboards to refresh within 500 ms under peak load, so that the data I act on is current.	Medium	8	2→3	In Progress
RTT-18	RTT-4	As an engineering manager, I want to be warned when approving a timesheet would push projected budget burn past 90%, so that I can act before the budget is blown.	Low	5	2→3	To Do

Backlog totals: **14 user stories, 52 story points**. By priority: 8 High (34 SP), 5 Medium (19 SP), 1 Low (5 SP). RTT-17 and RTT-18 rolled from Sprint 2 into a follow-up Sprint 3.



Story points, priority and status on the User Stories (Epic RTT-1 child items shown: RTT-5 = 3, RTT-6 = 2, RTT-8 = 3, RTT-7 = 5).

4. Story-point estimation (Fibonacci + Planning Poker)

Story points express the overall effort of a backlog item relative to work complexity, volume and uncertainty. Values are taken from the Fibonacci series (1, 2, 3, 5, 8, 13, …) — the non-linear gaps mirror the growing uncertainty of larger items and push the team to split work that is too big to estimate. Estimation used **Planning Poker**: for each story the team discussed it briefly, everyone privately chose a card, all cards were revealed together, and differences were discussed until the estimate converged.

SP	~ effort	Example story
2	~half a day	RTT-6 daily-cap rule, RTT-11 revise & resubmit
3	~one day	RTT-5 log time, RTT-9 submit timesheet, RTT-13 approve
5	~2-3 days	RTT-7 Jira link/sync, RTT-15 RBAC self-approval, RTT-16 burn-rate dashboard
8	~1 week	RTT-17 <500 ms dashboard performance under peak load (cross-cutting, high uncertainty)

Epic	Stories	Story points
RTT-1 Time Logging & Jira Integration	4	13
RTT-2 Timesheet Submission & Validation	3	8
RTT-3 Managerial Timesheet Approval	4	13
RTT-4 Project Budget Monitoring	3	18
Total	**14**	**52**

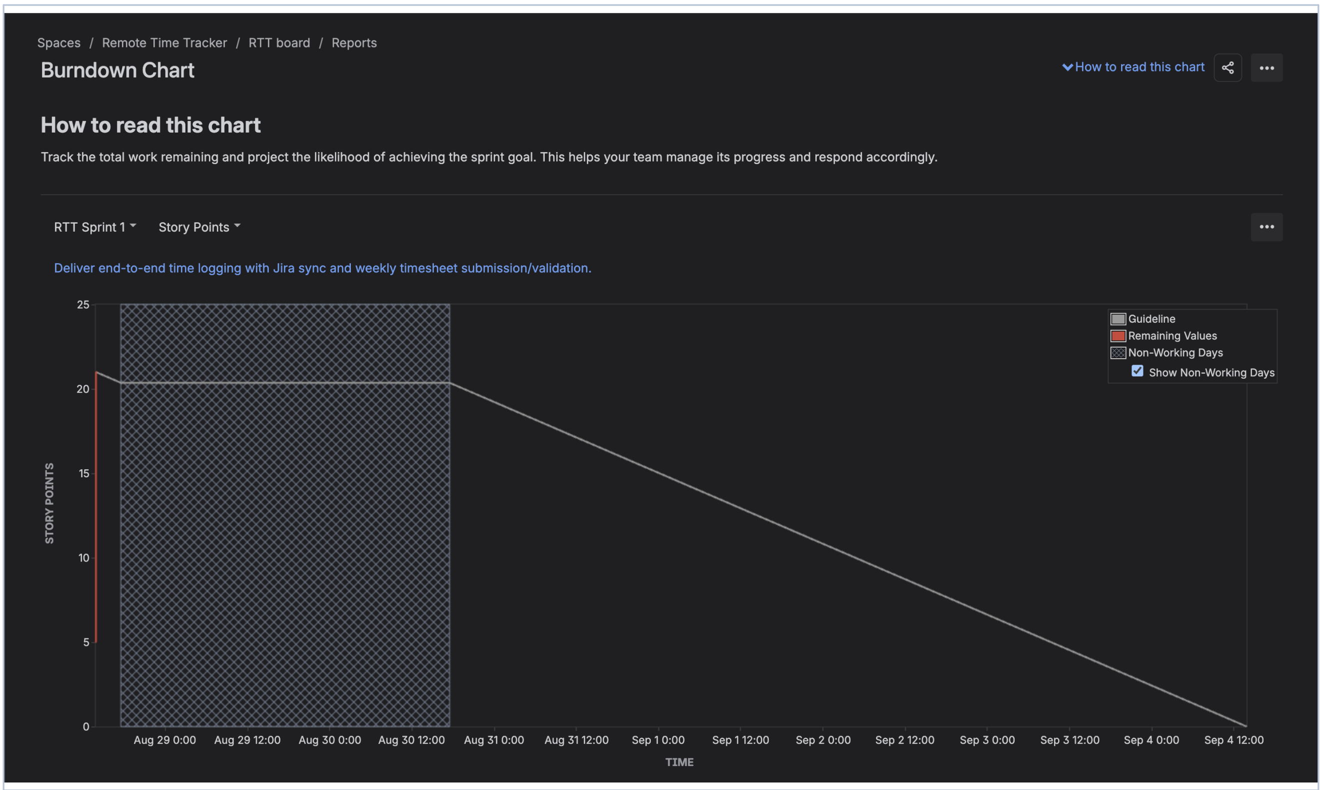
5. Backlog prioritisation

Items are ordered by business value and user need. The data-capture core (log time, Jira sync, submit timesheet) is **High** — nothing else works without it. Approval and the RBAC self-approval control are **High** — they are the compliance heart of the product. Validation niceties (daily cap, resubmit, entry validation) are **Medium**. Dashboard performance tuning is **Medium** and the conditional 90%-overrun warning is **Low** — valuable but deferrable. In the Jira backlog the stories are ranked in this order and grouped under their Epic.

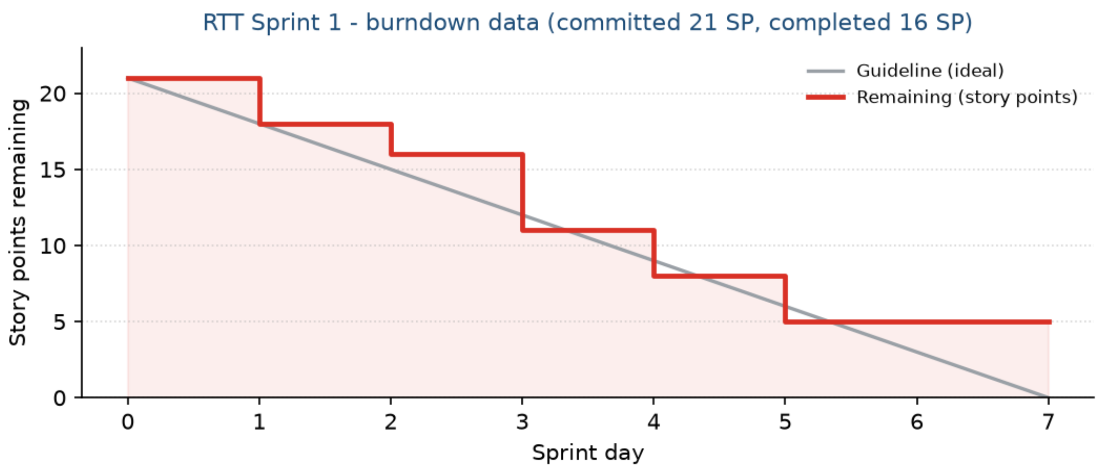
6. Sprint 1 simulation

Duration: 1 week. **Goal:** deliver end-to-end time logging with Jira sync and weekly timesheet submission / validation. **Scope:** Epics RTT-1 & RTT-2 — 7 stories, 21 story points (RTT-5..RTT-11).

Execution (cards dragged To Do → In Progress → Done on the Active-sprint board): RTT-5, RTT-6, RTT-7, RTT-8 and RTT-9 completed (16 SP). RTT-10 ended in “In Progress” and RTT-11 in “To Do”; both (5 SP) were carried into Sprint 2 on sprint completion.



Jira -> Reports -> Burndown Chart -> RTT Sprint 1. Grey = ideal guideline (21 -> 0); red = story points remaining. The remaining line ends above the guideline at 5 SP.



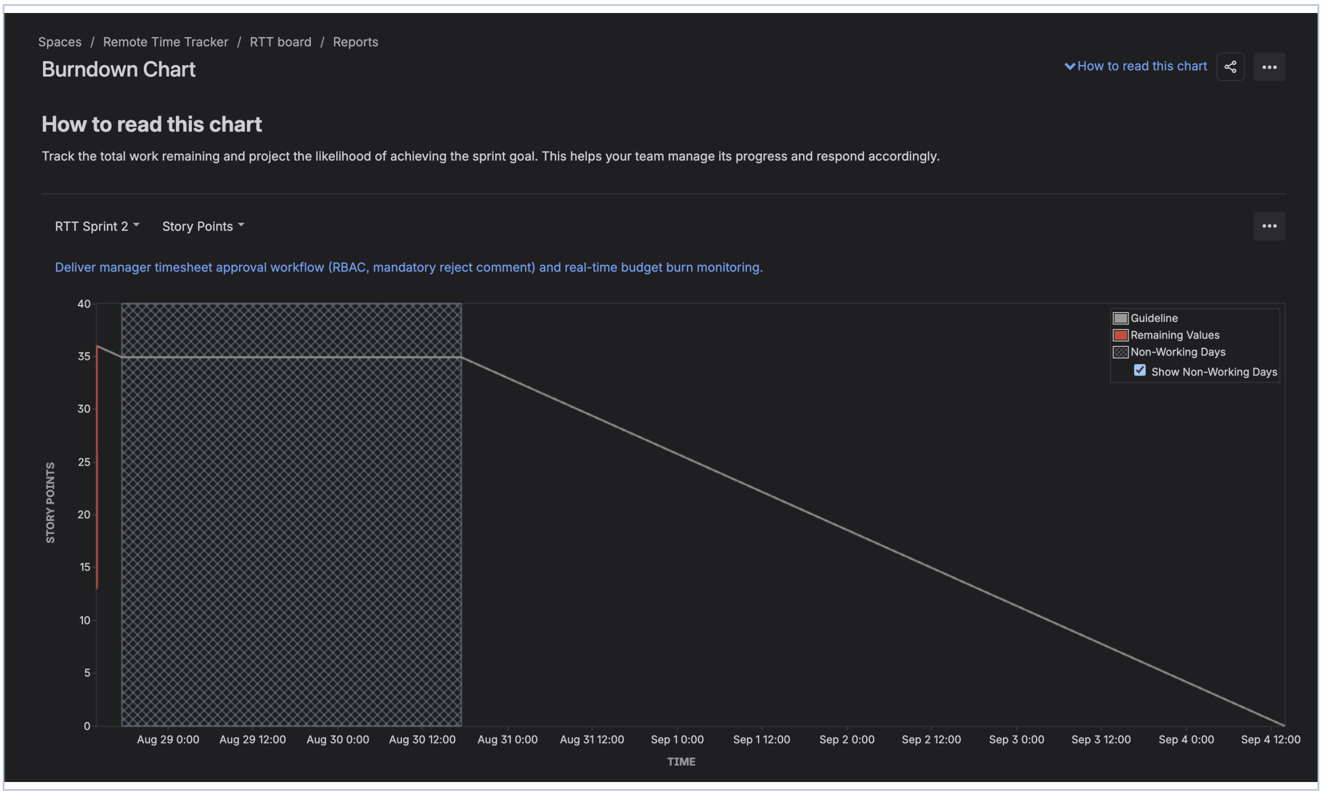
The same burndown data, plotted over the 1-week sprint for readability. Recorded events (Reports → Burndown Chart table): start 21 → 18 → 16 → 11 → 8 → 5 remaining at sprint end.

Sprint 1 result: 5 of 7 stories done, 16 / 21 SP delivered (76%). Velocity = 16.

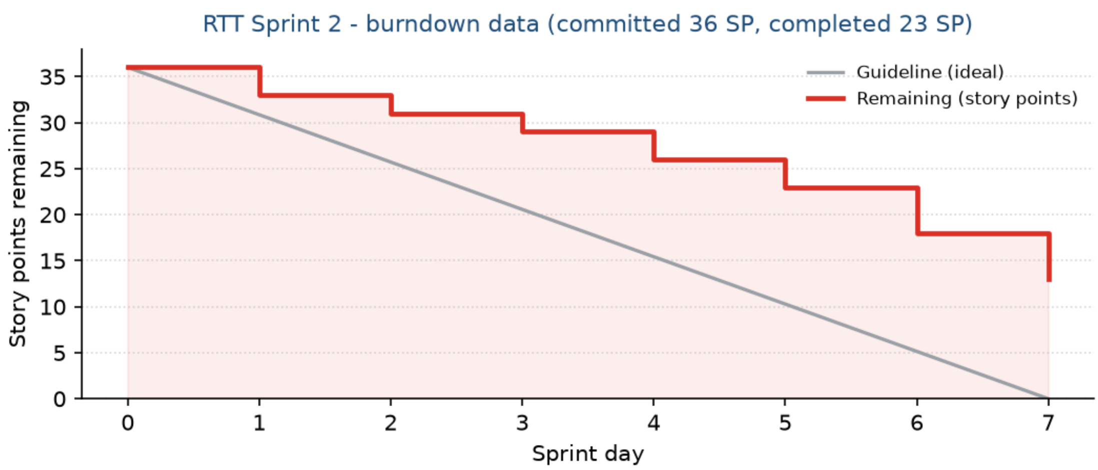
7. Sprint 2 simulation

Duration: 1 week. **Goal:** deliver the manager timesheet-approval workflow (RBAC, mandatory rejection comment) and real-time budget-burn monitoring. **Scope:** the two carry-over stories (RTT-10, RTT-11) plus Epics RTT-3 & RTT-4 — 9 stories, 36 story points.

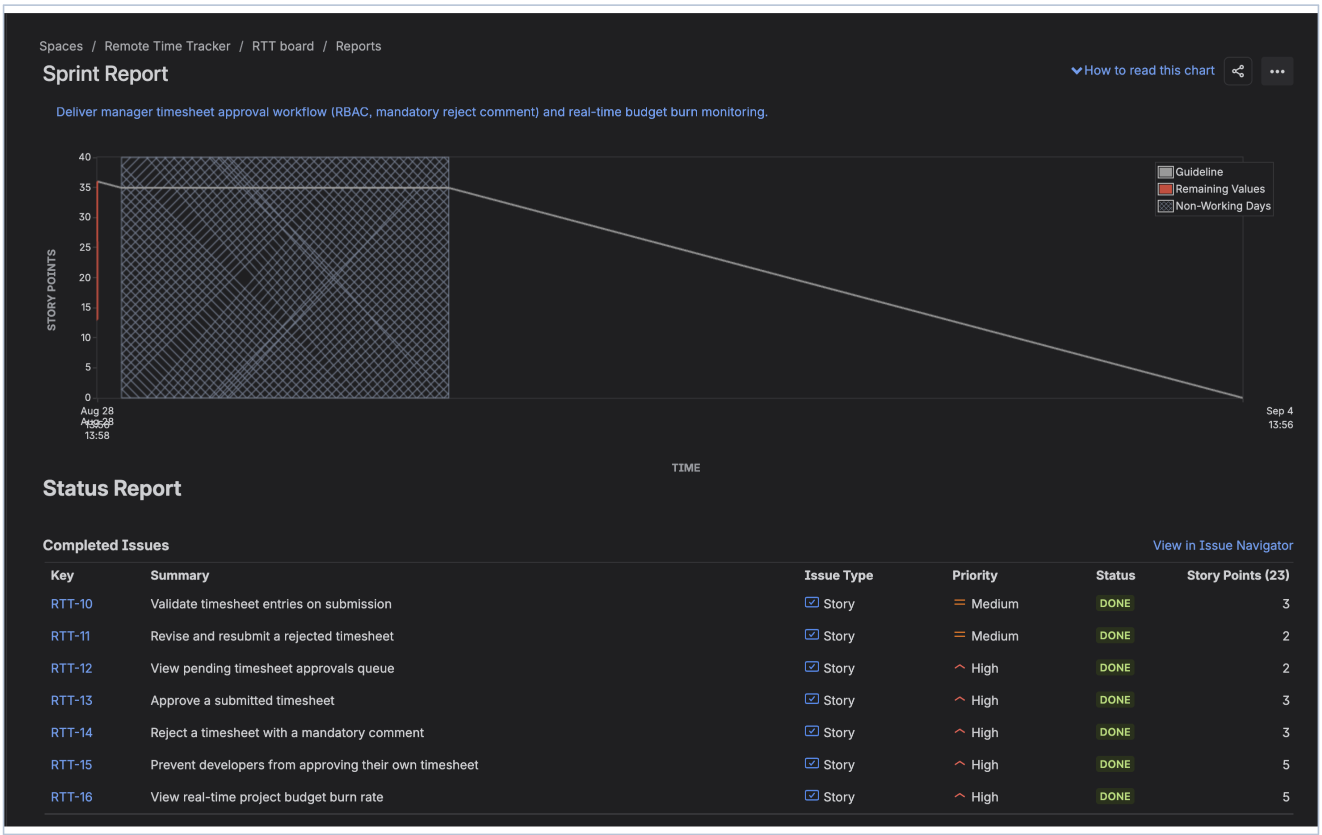
Execution: RTT-10, RTT-11, RTT-12, RTT-13, RTT-14, RTT-15 and RTT-16 completed (23 SP). RTT-17 (8 SP, the <500 ms performance story) ended in “In Progress” and RTT-18 (5 SP, overrun warning) in “To Do”; on completing the sprint both moved to a follow-up sprint.



Jira -> Reports -> Burndown Chart -> RTT Sprint 2.



The same data over the sprint: 36 → 33 → 31 → 29 → 26 → 23 → 18 → 13 remaining. The line again ends above the guideline (13 SP outstanding).



Spaces / Remote Time Tracker / RTT board / Reports

Sprint Report

How to read this chart

Deliver manager timesheet approval workflow (RBAC, mandatory reject comment) and real-time budget burn monitoring.

Status Report

Completed Issues

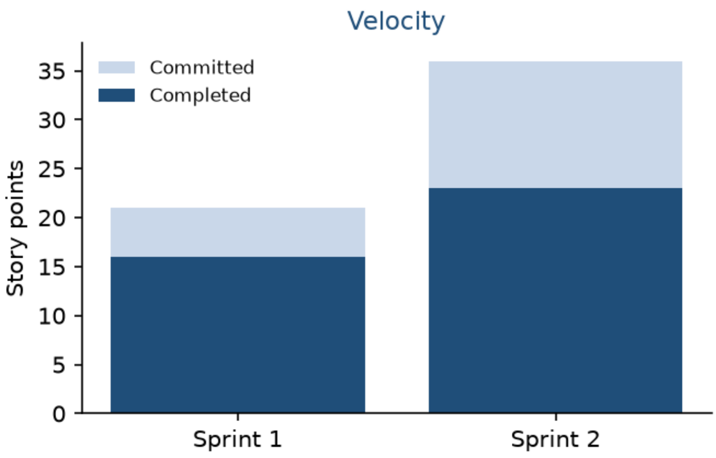
View in Issue Navigator

Key	Summary	Issue Type	Priority	Status	Story Points (23)
RTT-10	Validate timesheet entries on submission	Story	Medium	DONE	3
RTT-11	Revise and resubmit a rejected timesheet	Story	Medium	DONE	2
RTT-12	View pending timesheet approvals queue	Story	High	DONE	2
RTT-13	Approve a submitted timesheet	Story	High	DONE	3
RTT-14	Reject a timesheet with a mandatory comment	Story	High	DONE	3
RTT-15	Prevent developers from approving their own timesheet	Story	High	DONE	5
RTT-16	View real-time project budget burn rate	Story	High	DONE	5

Jira -> Reports -> Sprint Report for RTT Sprint 2 - Completed Issues total 23 story points (RTT-10..RTT-16), with priority and status per story.

Sprint 2 result: 7 of 9 stories done, 23 / 36 SP delivered (64%). Velocity = 23.

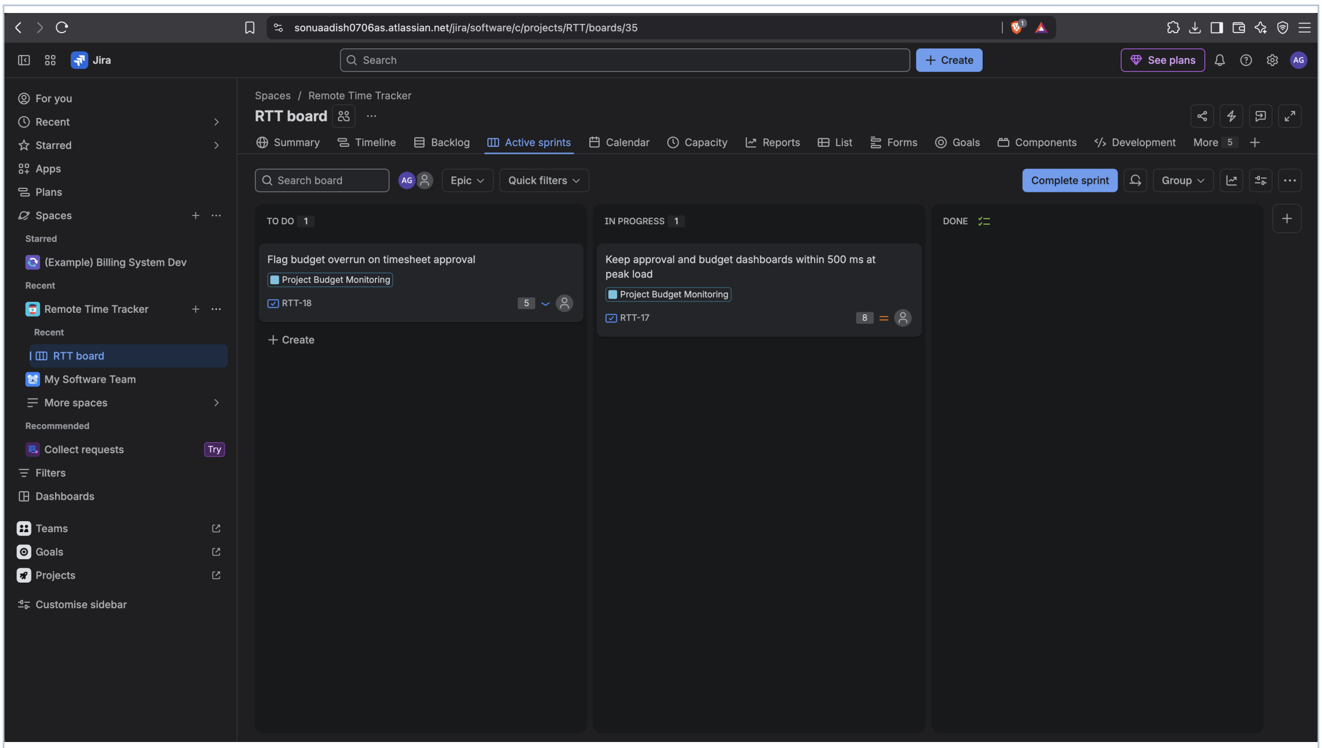
8. Velocity & project status



Committed vs completed story points per sprint (derived from the Sprint Reports above).

	Committed SP	Completed SP	Stories done	Velocity
Sprint 1	21	16	5 / 7	16
Sprint 2	36	23	7 / 9	23
Project	**52**	**39 (75%)**	**12 / 14**	**avg ~19.5**

Rolled into Sprint 3: **RTT-17** (dashboard <500 ms, 8 SP) and **RTT-18** (90% budget-overrun warning, 5 SP) — 13 SP.



Active-sprint Scrum board (RTT Sprint 3): the carry-over stories on the To Do / In Progress / Done board.

9. Reflection questions

1. Did your estimations reflect the actual effort?

Mostly. The 2/3-point stories (log time, submit, approve, reject-with-comment) behaved as expected. The two under-estimates were the cross-cutting items: RTT-15 (RBAC self-approval, estimated 5) touched auth on every approval path, and RTT-17 (<500 ms at peak, estimated 8) is really a performance / load-testing effort whose uncertainty a single 8 hid. In hindsight RTT-17 should have been split (instrument dashboards, add caching, load-test) so the uncertainty was visible.

2. Was your backlog well-prioritised?

Yes. Both sprints delivered a coherent, demoable slice: Sprint 1 gave a working log-time-and-submit path; Sprint 2 gave the manager approval workflow with the RBAC control. The two items that slipped were the lowest-value work in their epics (a performance NFR and a conditional Low-priority warning), so priority order protected the important scope.

3. How did your simulated sprint align with your plan?

Directionally yes, quantitatively no. The plan assumed both sprints would fully close. Sprint 1 finished 16 of 21 SP and Sprint 2 23 of 36 SP — the burndown lines stayed above the guideline in both. Sprint 2 was clearly over-committed: it carried 5 SP of Sprint 1 debt and then added a full 36 SP against a demonstrated velocity of only 16.

4. What insights did the burndown chart give about your team's capacity?

Sustainable capacity is about **16-20 story points per one-week sprint**, not the 36 planned for Sprint 2. A burndown that never touches the guideline and flattens near the end (as both sprints did) is the signature of over-commitment and of work that was not broken down finely enough. The fix for Sprint 3 was to take only the 13 SP of carry-over and split RTT-17.

Appendix — Jira workspace

Live workspace for instructor evaluation: **sonuaadish0706as.atlassian.net** → project **Remote Time Tracker (RTT)** → RTT board (id 35). Backlog (4 Epics, 14 stories grouped by Epic), story points on every story, completed Sprint 1 and Sprint 2 under Reports → Sprint Report, and Reports → Burndown Chart for both sprints.